Do Now · retrieval

Stacks and queues with arrays

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does scores[2] access in a C# array?

Q2

What is the difference between an array and a List<T> in C#?

Q3

What does the start pointer identify in a linked list?

Q4

For data = [10, 20, 30, 40], next = [2, -1, 1, 0] and start = 3, list the data values visited in order by following next pointers until -1.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Stack array — CODE:
int[] stack = new int[10];
int top = -1; — -1 means the stack is empty.

Core idea 2

Push and pop — CODE:
top++;
stack[top] = value;

value = stack[top];
top--; — Before push, reject top==stack.Length-1 (full); before pop, reject top==-1 (empty). Push then increments top; pop reads then decrements it.

Core idea 3

Queue array — CODE:
int[] queue = new int[10];
int front = 0;
int rear = -1; — rear tracks the newest item and front tracks the oldest.

Key visual · complete concept



New Learning · applying the ideas

Core idea 4

Enqueue and dequeue — CODE:
rear++;
queue[rear] = value;

value = queue[front];
front++; — This linear queue uses rear as the last occupied index. Reject enqueue when rear==queue.Length-1 and dequeue when front>rear. Freed earlier slots are not reused; a circular queue later adds wraparound.

Vocabulary · use the correct technical terms

stack — A LIFO data structure.

queue — A FIFO data structure.

LIFO — Last In, First Out.

FIFO — First In, First Out.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Trace a stack

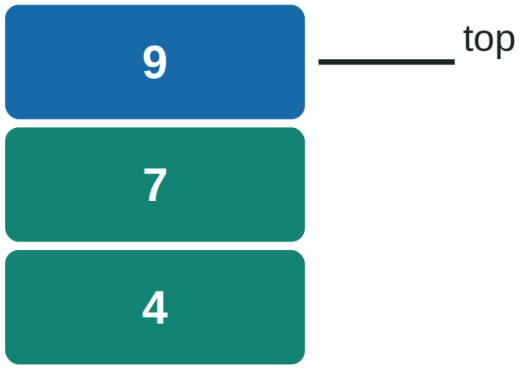
Follow the worked method

1. Store 8 — CODE:
top = 0
stack[0] = 8

2. Remove two values — CODE:
first = 7
second = 8
top = -1

3. Store result — CODE:
top = 0
stack[0] = 15

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Complete the same state
CODE:
Start: empty
push 8 → [8]
push 7 → [?]
A: pop two, add, push → [?]

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Implement push/pop

Create an integer stack array and write code that pushes 5, 8 and 3, then pops and prints one item.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Explain an error

Using the OCR 2023 calculator-stack context, explain what happens if an operation tries to pop more values than are stored.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A stack removes the newest stored item first, while a queue removes the oldest stored item first.

Use key vocabulary from memory

1. Explain stack in your own words.

2. Explain queue in your own words.

3. Explain LIFO in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.